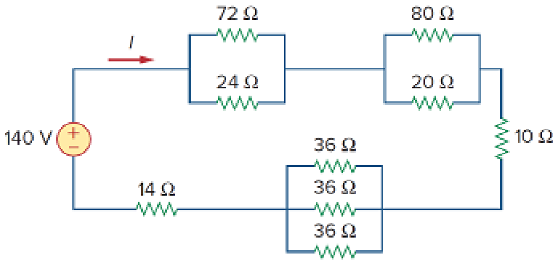
Chapter 2, Problem 46P

Problem

Find *I* in the circuit of Fig. 2.110.

Figure 2.110:



Step-by-step solution

Step 1 of 3

Consider the circuit of textbook Figure 2.110. The resistors $72\ \Omega$ and $24\ \Omega$ are in parallel. Therefore their equivalent resistance is,

$$\begin{aligned}\left(R_{72\Omega} \parallel R_{24\Omega}\right) &= \frac{(72)(24)}{96} \\ &= 18\ \Omega\end{aligned}$$

The resistors $80\ \Omega$ and $20\ \Omega$ are in parallel. Therefore their equivalent resistance is,

$$\begin{aligned}\left(R_{80\Omega} \parallel R_{20\Omega}\right) &= \frac{(80)(20)}{100} \\ &= 16\ \Omega\end{aligned}$$

The resistors $36\ \Omega$, $36\ \Omega$ and $36\ \Omega$ are in parallel. Therefore their equivalent resistance is,

$$\begin{aligned}\left(R_{36\Omega} \parallel R_{36\Omega} \parallel R_{36\Omega}\right) &= \frac{R_{36\Omega}}{3} \\ &= \frac{36}{3} \\ &= 12\ \Omega\end{aligned}$$

Step 2 of 3

The resultant circuit diagram is as shown in Figure 1.

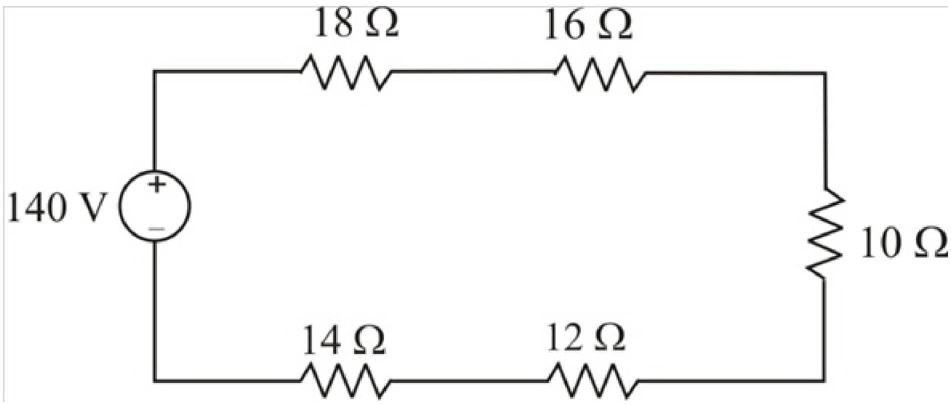


Figure 1

Step 3 of 3

The total resistance of the circuit in Figure 1 is,

$$\begin{aligned} R &= 18+16+10+12+14 \\ &= 70\ \Omega \end{aligned}$$

The current through the above circuit is,

$$\begin{aligned} I &= \frac{V}{R} \\ &= \frac{140}{70} \\ &= 2\ \text{A} \end{aligned}$$

Therefore, the current through the circuit of textbook Figure 2.110 is $\boxed{2\ \text{A}}$.